

CoinWise AI — Advanced Fintech Synthesis

AI-driven, alternative-data, VND-localized learn-to-trade platform for the Vietnamese market

Project Report (5-page submission) · **AI Integration & API Refinement** | For: UII Incubator — Demo Day | Stage: deployed MVP (paper-trading)

1. Executive Summary & Product Concept

CoinWise AI is a learn-by-doing investing platform for Vietnam's retail crypto market — among the world's highest in adoption (consistently top-ranked by Chainalysis; ~17–20M holders) yet served only by English-/USD-first tools that offer generic charts and no localized decision support. Users are highly active but under-educated and over-exposed to FOMO and scams. We close that gap by combining four things in one product:

- **(1) Academy** — structured courses on trading and on how to use AI in crypto;
- **(2) Risk-free paper trading** — 1,000,000 USDT of virtual capital to practise real strategies with zero downside;
- **(3) AI copilot + alternative-data engine** — proprietary signals and an agentic chatbot that explain *why* to act;
- **(4) Arena** — skill-based tournaments with a VND entry fee and VND prize pool, turning learning into a social loop.

The thesis: people learn to trade by practising safely with feedback, then competing — building real skill before risking real money. Beneath this consumer wrapper sits the graded core of this assignment: a **proprietary alternative-data AI engine**, a **custom VND-localized OpenAPI server** that fixes the currency gaps global providers (Stripe/Plaid) leave open, and an **agentic Gemini assistant** — all live in the deployed MVP today. Indicative sizing (to validate): TAM ~17–20M, SAM ~3–4M, SOM (3-yr) ~150k–300k users.

2. AI Methodology — Alternative Data (Part A · 30%)

Alternative data is any non-traditional signal beyond price/transaction history — public social & news text, community voting, market-psychology indices. The in-app **Alt-Data Lab** exposes the full chain transparently: *real data* → *AI analysis* → *a concrete fintech action*. Sources are live, free, and need no paid keys:

Source	What it provides	Role
Hacker News (Algolia)	Story titles + upvote scores per coin	Primary text corpus
Reddit (r/CryptoCurrency...)	Post titles/bodies + upvotes	Supplementary text
alternative.me Fear & Greed	Market psychology index 0–100	Fusion signal
CoinGecko community	Up/down vote %, dev/community score	Fusion signal

The pipeline (GET /api/v1/ai/alt-data/pipeline/:symbol) runs Collect → Analyse → Fuse → Apply. Four course techniques run on one weighted corpus: **VADER lexicon** (explainable per word), a **Multinomial Naive Bayes** we trained, **Z-score anomaly detection** (mention-volume spikes = early FOMO), and **multi-source fusion** into a single composite score, confidence and signal — auto-reweighting when a source is blocked (e.g. Reddit 403), with LIVE/HYBRID/DEMO provenance on every datapoint. Nothing is a black box.

The trained model uses **distant supervision**: ~289 hand-labeled “gold” + ~500 auto-labeled “silver” documents, tested on gold only so accuracy can't be inflated — reaching **accuracy ≈ 73.6% / macro-F1 ≈ 73.9% (vocab 2,268)**. It is trained in a reproducible Python notebook, then its weights are exported to TypeScript so it *serves inside a serverless function with no Python at runtime* — identical predictions, production latency.

Why this is our competitive advantage. Signals feed two products — **Fraud Shield** (flags FOMO/momentum-chasing orders) and **AI Advisor** (±25% allocation tilt). The moat is a **data flywheel**: Community Pulse scores every user comment with our own model, growing a Vietnam-specific labeled dataset that compounds with usage and cannot be bought off-the-shelf or replicated by a global incumbent — giving us three structural edges: explainability (defensible token-by-token), resilience (graceful degradation + provenance), and locality (a signal tuned to Vietnam).

3. Server Architecture — Custom OpenAPI Server (Part B · 30%)

Global sandboxes (Stripe, Plaid) don't process VND natively, don't integrate with local banking rails, and can't serve our AI analytics in a structured contract. So we built our **own OpenAPI server** (Hono + TypeScript), documented with Swagger UI at /docs plus a

machine-readable **openapi.yaml**, deployed as a Vercel serverless function beside the React/Vite frontend — the single source of truth for paper-account state.

- **VND money layer** — /fx/rates & /fx/convert expose live USD↔VND (~26,300 VND/USD), so every balance, price and fee is correct in local currency.
- **CoinWise Bank** — a custom local banking rail (/bank/...) processing deposits, withdrawals, subscription billing and Arena entry fees & prize payouts *in VND*: exactly the “replace the global sandbox with a localized backend” the rubric rewards.

Endpoint	Purpose
GET /ai/alt-data/pipeline/:symbol	Full alt-data → AI signal pipeline for a coin
POST /ai/alt-data/classify	Score any text with the trained model (live)
GET /ai/alt-data/model/info	Model metadata + evaluation metrics
POST /ai/fraud-check-real	Alt-data-driven transaction risk
POST /agent/execute	Tool dispatcher consumed by the AI agent
GET /fx/rates · /bank/...	VND localization + local banking rail

4. Agentic AI & Localized UX (Part C · 20%)

Beyond chat to action. The assistant is a **Gemini function-calling agent** that calls our OpenAPI tools (getPortfolio, getSentiment, getAIInsight, convertCurrency, placeTrade) to actually do things. It understands localized intent — type “*mua giúp tôi 5.000.000đ BTC*” and it converts VND→USD, quotes the trade, asks for confirmation, then executes through the server, every tool call shown transparently. A global USD↔VND toggle re-renders all amounts, alongside live candlestick charts with exchange-grade indicators.

The learn-to-trade loop: Academy → practise on the live market with 1,000,000 USDT virtual capital and AI feedback (Fraud Shield + AI Advisor explain each move) → Arena tournaments with VND entry/prize via CoinWise Bank. This learn-practise-compete loop drives retention today and produces the labeled engagement data that compounds the moat tomorrow.

5. Updated Roadmap & Incubation Readiness (Part D · 20%)

Why the AI makes this incubation-viable for UII: a **data moat** that compounds with usage, **real localized infrastructure** no global API offers in VND, and an **education-first paper-trading on-ramp** with a low regulatory surface that de-risks a future real-money launch. Revenue lines are already in the product: subscriptions (Pro \$19/mo ~500k^d, Elite \$99/mo), Arena rake (entry \$5–\$500), Earn yield (APY 2.5–18%), and future brokerage spread / B2B data API.

Phase	Milestones
0 — Now (done)	Deployed AI MVP: alt-data engine, VND OpenAPI server, agentic AI, data-flywheel foundation
1 — Incubation (0–6 mo)	eKYC/compliance design; validate conversion & ARPU; expand alt-data (on-chain, app-usage); harden infra
2 — Real-money beta (6–12 mo)	Local banking rail (VietQR/Napas); licensed/partnered brokerage; paid subscriptions live
3 — Scale (12–24 mo)	Grow to SOM; B2B data/signal API; new asset classes; SEA expansion

Unit economics (to validate): 3–5% free→paid, blended paid ARPU ~\$30/mo, target LTV/CAC > 3x. **Key risks** — regulatory uncertainty, free-source reliability, and ~74% model accuracy — are mitigated by the education-first launch (low regulatory surface), provenance + graceful degradation, and continuous retraining via the data flywheel. **The ask:** UII acceptance + pre-seed for eKYC/compliance, real banking integration, ML expansion and go-to-market. The project answers all four rubric bands — alt-data & moat (30%), localized infrastructure (30%), agentic AI & UX (20%), incubation readiness (20%) — with a working MVP behind every claim.